Matter-Antimatter Asymmetry and the Information Writing Constraint:

The Baryon Asymmetry as a Self-Consistency Condition of the Informational Actualization Model

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Abstract

The matter-antimatter asymmetry of the universe, one surviving baryon per approximately 10^9 annihilation pairs, is conventionally treated as a free parameter of the standard model, explained qualitatively by the Sakharov conditions but not derived from first principles. Within the Informational Actualization Model (IAM), in which the cosmological constant accumulates as the Landauer cost of irreversible gravitational decoherence on baryonic timelike worldlines, we ask whether the baryon asymmetry $\eta \approx 6 \times 10^{-10}$ is genuinely free or whether it is a self-consistency condition of the universe's information writing constraint.

The IAM framework requires that the rate at which baryonic matter writes classical information onto the cosmic horizon remain compatible with the horizon's accommodation capacity over the full cosmic history. We identify a self-consistency loop in which matter content sets the decoherence rate, which sets the information writing rate, which must match the horizon accommodation rate, which feeds back to constrain the matter content. We propose that $\eta \approx 6 \times 10^{-10}$ is the fixed point of this loop rather than an arbitrary output of the Sakharov mechanism.

A further consequence follows from IAM's identification of dark matter as the geometric potential half and dark energy as the kinetic informational half of the cosmological virial partition [Mahaffey, 2026c]. The 10^9 annihilation partners per surviving baryon are not missing. They constitute the dark sector. The universe retains the complete record of its annihilation history in geometric form.

We do not claim a quantitative derivation of η from the present framework. We identify the physical conditions a derivation would need to satisfy, present the self-consistency loop as a concrete research target, and propose a falsification test: allowing η to float as a free parameter in the IAM MCMC validation infrastructure and asking whether the posterior returns $\eta \approx 6\times10^{-10}$ without external constraint.

Keywords: baryon asymmetry; matter-antimatter asymmetry; Sakharov conditions; gravitational decoherence; Landauer's principle; holographic principle; dark matter; dark energy; Informational Actualization Model

1 Introduction

The observed universe contains matter and no appreciable antimatter. The baryon-to-photon ratio,

$$\eta \equiv \frac{n_b - n_{\bar{b}}}{n_\gamma} \approx 6.1 \times 10^{-10}, \tag{1}$$

is measured independently from Big Bang nucleosynthesis [Cyburt et al., 2016] and from the cosmic microwave background power spectrum [Planck Collaboration, 2020], with consistent results. The two measurements together confirm that the asymmetry was established before nucleosynthesis and is a genuine property of the early universe rather than a local accident.

The Sakharov conditions [Sakharov, 1967] identify the necessary ingredients for generating a baryon asymmetry from a symmetric initial state: baryon number violation, C and CP violation, and departure from thermal equilibrium. The standard model satisfies all three conditions in principle, though the magnitude of CP violation in the CKM matrix [Kobayashi & Maskawa, 1973] appears insufficient to generate the observed η without extensions to the standard model. This has motivated a large literature on baryogenesis mechanisms including electroweak baryogenesis, leptogenesis, and Affleck-Dine baryogenesis [Rubakov & Shaposhnikov, 1996, Davidson et al., 2008].

What these mechanisms have in common is that they treat η as an output of dynamical processes in the early universe, with its precise value depending on parameters that are not themselves derived from a deeper principle. In this sense η remains a free parameter: the standard model and its extensions can accommodate a wide range of values, and the observed value $\eta \approx 6 \times 10^{-10}$ is not understood as necessary rather than contingent.

The present paper asks a different question. Within the Informational Actualization Model [Mahaffey, 2026a], the universe has an information writing budget: baryonic matter generates irreversible classical information through gravitational decoherence on timelike worldlines, and this information is encoded on the cosmic horizon at a cost set by Landauer's principle [Landauer, 1961]. The total accumulated cost over cosmic history is identified with the cosmological constant [Mahaffey, 2026d]. This writing process has a natural rate and a natural capacity. We ask whether these two quantities together constrain the fraction of matter that must survive matter-antimatter annihilation in order for the universe to write its own history coherently.

2 The IAM Framework

The Informational Actualization Model [Mahaffey, 2026a] extends the thermodynamic derivation of the gravitational field equations [Jacobson, 1995] by including an informational contribution to the horizon entropy functional. The extension rests on three established physical principles.

Gravitational decoherence on timelike worldlines is the dominant source of irreversible classical information production in the universe. Every massive particle interacting gravitationally undergoes continuous decoherence, producing classical information at a rate set by the virial theorem [Mahaffey, 2026b].

Landauer's principle [Landauer, 1961] establishes that each irreversible bit of classical information produced at the cosmic horizon temperature $T_H = \hbar H_0/(2\pi k_B)$ [Gibbons & Hawking, 1977] carries a minimum thermodynamic cost $E_{\text{bit}} = \hbar H_0 \ln 2/(2\pi)$.

The holographic principle ['t Hooft, 1993, Susskind, 1995] requires that irreversible classical information produced by gravitational decoherence be encoded on the cosmic horizon. The maximum information content is set by the Bekenstein-Hawking entropy [Bekenstein, 1973, Hawking, 1975]: $S = A_H/(4l_P^2)$, where l_P is the Planck length.

The result is a sector-dependent modification of the perturbation equations: matter experiences a Hubble friction enhancement $\mu(a) < 1$ while photon-sector observables retain $\Sigma(a) = 1$ exactly. The coupling constant $\beta_m = \Omega_m/2$ is derived from the virial theorem and confirmed independently by 17 converged Planck 2018 MCMC chains [Mahaffey, 2026a].

Within this framework, dark matter is identified as the accumulated geometric potential half of prior gravitational decoherence events, and dark energy is the kinetic informational half [Mahaffey, 2026c]. These are not two independent phenomena. They are two expressions of the virial equilibrium $2K + V = 0$ operating at cosmological scales. The virial ratio $\Omega_m/[\beta_m \cdot E(a)]$ asymptotes to exactly 2.0 at $a = 1$, identifying the present epoch as the moment of first cosmological virialisation. The coincidence problem is thereby dissolved.

3 The Information Writing Constraint

Within IAM, the cosmological constant at any epoch a accumulates as:

$$\frac{\Lambda(a)}{\rho_{\mathrm{vac}}} \propto \int_{a_{\mathrm{EW}}}^{a} \frac{\Omega_b(a')/\Omega_{\mathrm{tot}}(a')}{l_H(a')^2/l_P^2} \frac{da'}{a'^2 H(a')/H_0} \tag{2}$$

where $\Omega_b(a')/\Omega_{\mathrm{tot}}(a')$ is the baryonic fraction of the total energy density at scale factor a', and $l_H(a') = c/H(a')$ is the Hubble length at that epoch [Mahaffey, 2026d]. The integration begins at the electroweak transition $a_{\mathrm{EW}} \approx 2.3 \times 10^{-15}$, which IAM identifies as the onset of the matter sector and the origin of duration [Mahaffey, 2026e].

The integrand in equation (2) is the information writing rate per unit scale factor: the fraction of the horizon's Planck-area bits being actualized by baryonic decoherence at each epoch, weighted by the baryonic fraction of energy and the inverse square of the Hubble length. This rate has two natural limits.

Upper limit: the writing rate cannot exceed the horizon accommodation rate. The horizon can encode at most $N_H = A_H/(4l_P^2)$ bits at any epoch. If baryonic decoherence attempts to write information faster than the horizon can accommodate it, the excess cannot be encoded. In physical terms, this corresponds to a universe with too much matter: structure collapses before the horizon has grown sufficiently to accommodate the encoded history.

Lower limit: the writing rate must be sufficient to actualize the observed cosmological constant over 13.8 billion years. If baryonic matter is too dilute, too few decoherence events occur per unit time, and the accumulated Landauer cost falls below the observed Λ. In physical terms, this corresponds to a universe with too little matter: the writing rate is insufficient and the dark sector fails to accumulate the observed energy density.

These two limits together constitute a constraint on the baryonic matter density. Within the IAM framework, the baryonic fraction $\Omega_b/\Omega_m = 0.1564$ [Planck Collaboration, 2020] is not a free input to the cosmological constant calculation. It was derived from IAM's physical identification of dark matter as accumulated geometry [Mahaffey, 2026c] and applied to the calculation independently of numerical agreement [Mahaffey, 2026d].

4 The Self-Consistency Loop

We now identify the connection between the information writing constraint and the baryon asymmetry. The argument proceeds in the following logical steps.

Step 1. The matter content of the universe is set at the QCD transition ($T \approx 150$ MeV, $a_{\mathrm{QCD}} \approx 1.6 \times 10^{-12}$) by the baryon asymmetry η. For every 10^9 photons in the thermal bath, $\eta \times 10^9$ baryons survive annihilation. The surviving baryon density is therefore:

$$n_b = \eta \cdot n_\gamma \tag{3}$$

Step 2. The surviving baryonic matter density determines the decoherence rate. Within IAM, baryons on timelike worldlines generate irreversible classical information through gravitational interactions at a rate proportional to their number density and the local gravitational potential, set by the virial theorem [Mahaffey, 2026b].

Step 3. The decoherence rate sets the information writing rate onto the cosmic horizon. The accumulated writing over cosmic history is governed by equation (2), with Ω_b determined by η through equation (3).

Step 4. The information writing rate must remain compatible with the horizon accommodation rate at every epoch. The horizon accommodation rate is set by the Hubble expansion, which is itself governed by the total matter-energy content including the dark sector. The dark sector is identified within IAM as the accumulated record of prior decoherence events [Mahaffey, 2026c].

Step 5. The dark sector feeds back to the matter content through the virial partition. The geometric potential half of each decoherence event contributes to dark matter; the kinetic informational half contributes to dark energy. The total energy budget therefore depends on the full history of baryonic decoherence, which depends on η through Steps 1 through 4.

This closes the loop:

$$\eta \longrightarrow n_b \longrightarrow \dot{W}_{\text{info}} \longrightarrow \Lambda_{\text{acc}} \longrightarrow \{\rho_{\text{dm}}, \rho_{\text{de}}\} \longrightarrow H(a) \longrightarrow \eta \tag{4}$$

We propose that the observed value $\eta \approx 6 \times 10^{-10}$ is the fixed point of this loop: the unique value of the baryon asymmetry consistent with a universe that can write its own decoherence history onto a horizon of finite capacity over a finite cosmic time.

This is not a claim that the Sakharov conditions are unnecessary. The Sakharov conditions provide the dynamical mechanism by which a baryon asymmetry is generated. The present proposal is that among the values the Sakharov mechanism could in principle produce, the universe selects the value consistent with the self-consistency loop (4).

5 The Dark Sector as Encoded Annihilation History

A direct consequence of IAM's dark sector identification is a reinterpretation of the matter-antimatter annihilation itself.

In the standard picture, the $10^9 - 1$ matter-antimatter pairs that annihilate per surviving baryon produce photons, which join the thermal bath and are observed today as the CMB. The annihilation products are accounted for energetically but the geometric record of the annihilation events is not separately tracked.

Within IAM, every gravitational decoherence event deposits two contributions onto the cosmic horizon: the geometric potential half, which accumulates as dark matter, and the kinetic informational half, which accumulates as dark energy [Mahaffey, 2026c]. The matter-antimatter annihilation events at the QCD transition constitute an enormous and sudden burst of decoherence activity. The geometric record of those events is not lost. It is encoded on the horizon and observed today as the dark sector.

The numerical consequence is notable. The observed ratio of dark matter plus dark energy to baryonic matter is approximately:

$$\frac{\rho_{\mathrm{dm}} + \rho_{\mathrm{de}}}{\rho_b} \approx \frac{0.265 + 0.685}{0.049} \approx 19.4 \tag{5}$$

using Planck 2018 values [Planck Collaboration, 2020]. If the dark sector is the encoded record of the annihilation history, then approximately 10^9 annihilation events per surviving baryon deposited their geometric record onto the horizon. The ratio in equation (5) reflects the writing efficiency of the annihilation epoch, not a coincidental energy balance.

We note that this reinterpretation does not modify the standard energy accounting. The photons produced by annihilation carry the kinetic energy of the process; IAM identifies the geometric potential energy as separately encoded. These are complementary descriptions of the same events at different levels of physical description.

The conclusion is a restatement of IAM's central identification in the context of the baryon asymmetry: the universe is not 95% dark and 5% visible because dark matter and dark energy are mysterious substances. The universe is 95% memory and 5% present. The dark sector is the accumulated record of what has already been irreversibly actualized, beginning with the annihilation history at the QCD transition.

6 The CP Violation Connection

The Sakharov conditions require CP violation as a necessary ingredient for generating a baryon asymmetry. Within IAM, the weak force occupies a structurally distinct position from the other three forces.

The three forces governed by the Aristotelian Principle [Mahaffey, 2026f] — gravity, electromagnetism, and the strong nuclear force — satisfy the equilibrium condition between potential and actual at every scale at which they have been tested. The virial theorem $2K + V = 0$ holds for gravitational and Coulomb interactions from atomic scales ($\sim 10^{-11}$ m) to the cosmic horizon ($\sim 10^{26}$ m) [Mahaffey, 2026b]. These forces are forces of being: they maintain equilibrium between realized and unrealized states.

The weak force is the structural exception. It violates CP symmetry, drives the arrow of time, and produces irreversible transitions. It is the force of becoming: it does not maintain equilibrium but drives the system from one state to another without the possibility of reversal. Electroweak symmetry breaking at $T \approx 100$ GeV is identified within IAM as the first irreversible act of the universe — the origin of duration itself [Mahaffey, 2026e].

CP violation and the arrow of time are therefore the same asymmetry expressed at different scales. A universe in which the IAM activation function $E(a) = \exp(1-1/a)$ runs monotonically forward requires CP violation as a structural necessity, not a contingent feature of the standard model. The question whether the magnitude of CP violation in the CKM matrix [Kobayashi & Maskawa, 1973] is constrained by the self-consistency condition (4) is identified here as an open problem. A universe that requires $\eta \approx 6 \times 10^{-10}$ as its fixed point may also require a specific value of the CKM CP phase δ to generate precisely that asymmetry through the Sakharov mechanism. This connection is not derived here but is identified as a candidate direction for future work.

7 What a Derivation Would Require

The self-consistency loop identified in Section 4 is a physical argument, not a completed derivation. We identify here the specific elements a rigorous derivation of η from IAM would require, in order to make the research target precise.

Element 1: A quantitative writing rate at the QCD transition. The information writing rate in equation (2) is formulated for the post-recombination universe where baryons dominate the matter sector. A derivation of η requires extending this rate to the QCD epoch ($a \approx 1.6 \times 10^{-12}$), where the universe is radiation-dominated and the baryon fraction is a small perturbation on the thermal bath. The form of the writing rate integrand in this regime is not established within the current IAM framework and constitutes the primary technical obstacle.

Element 2: A horizon accommodation capacity at the QCD transition. The fixed point condition requires comparing the writing rate to the horizon accommodation capacity at the epoch when η is set. At a_{QCD}, the Hubble length is $l_H \approx 10^{-3}$ pc and the horizon contains approximately 10^{40} Planck-area bits. Whether the writing rate at this epoch saturates, undershoots, or overshoots this capacity is the quantitative question that determines η.

Element 3: A connection between the writing constraint and the Sakharov mechanism. The self-consistency loop constrains the *result* of baryogenesis rather than its mechanism. A complete derivation would identify how the information writing constraint interfaces with the dynamical Sakharov conditions to select the fixed point $\eta \approx 6 \times 10^{-10}$. This likely requires a treatment of the electroweak phase transition within the IAM framework beyond what is currently developed.

These three elements are identified as the research program implied by the present paper. The loop in equation (4) is the structural claim. The derivation of its fixed point is the open problem.

8 A Proposed Falsification Test

In the absence of a completed analytical derivation, a numerical test is available within the existing IAM validation infrastructure [Mahaffey, 2026a].

The IAM MCMC validation uses Cobaya [Torrado & Lewis, 2021] with MGCAMB [Wang et al., 2023] and the Planck 2018 likelihoods [Planck Collaboration, 2020]. The baryon density $\Omega_b h^2$ is a standard free parameter in these chains. The baryon-to-photon ratio η is related to Ω_b by:

$$\eta = \frac{\Omega_b \rho_c}{\langle m_b \rangle n_\gamma} \approx 2.74 \times 10^{-8}\, \Omega_b h^2 \tag{6}$$

where $\langle m_b \rangle$ is the mean baryon mass and n_γ is the photon number density today [Cyburt et al., 2016].

The proposed test proceeds as follows. A dedicated IAM chain is run with $\Omega_b h^2$ freed from its Planck prior and the IAM self-consistency condition applied as a derived constraint rather than an input. If the posterior of $\Omega_b h^2$ — and therefore of η through equation (6) — returns a value consistent with the observed $\eta \approx 6 \times 10^{-10}$ without external constraint, this would constitute numerical evidence that the baryon asymmetry is selected by the IAM information writing framework rather than being a free parameter.

This test is falsifiable in both directions. If the unconstrained posterior returns $\eta \approx 6 \times 10^{-10}$, the self-consistency loop is supported. If the posterior returns a significantly different value, the loop as currently formulated does not select the observed baryon asymmetry and the proposal requires revision or rejection.

The chain has not yet been run. The test is proposed here as a concrete, executable next step in the IAM validation program, with the prediction stated in advance of the result.

9 Relation to Prior Work

Baryogenesis has been approached from many directions. Electroweak baryogenesis [Rubakov & Shaposhnikov, 1996] generates η through sphaleron processes at the electroweak phase transition. Leptogenesis [Davidson et al., 2008] produces a lepton asymmetry via heavy neutrino decay which is subsequently converted to a baryon asymmetry. Affleck-Dine baryogenesis uses flat directions in supersymmetric potentials. In each case, η is an output of specific dynamics whose parameters are not themselves derived.

The present approach differs in that it does not propose a new dynamical mechanism for baryogenesis. The Sakharov conditions and their realizations in the standard model and its extensions are accepted as the correct description of how the asymmetry was generated. The proposal is that the magnitude of the asymmetry is not set by the dynamics alone but is further constrained by the requirement that the resulting baryonic matter density be consistent with the universe's information writing capacity.

The closest conceptual parallel is the anthropic reasoning of Weinberg [Weinberg, 1989], which constrains the cosmological constant by requiring structure formation. The present approach differs in that it does not invoke observer selection. The information writing constraint is a physical requirement of the IAM framework, not a selection criterion imposed from outside. The universe is not required to permit observers. It is required to write its own history coherently.

10 Conclusions

We have identified a self-consistency loop within the Informational Actualization Model that connects the baryon asymmetry η to the universe's information writing constraint. The loop requires that the rate at which baryonic matter writes classical information onto the cosmic horizon remain compatible with the horizon's accommodation capacity over the full cosmic history. We propose that the observed value $\eta \approx 6 \times 10^{-10}$ is the fixed point of this loop.

The 999,999,999 annihilation partners per surviving baryon are not absent from the physical record. They are encoded on the cosmic horizon as the dark sector. Dark matter is the geometric potential half and dark energy is the kinetic informational half of the accumulated annihilation history [Mahaffey, 2026c]. The universe retains a complete geometric record of the asymmetry that made its existence possible.

We do not claim a quantitative derivation of η. We identify the three elements such a derivation would require, and we propose a falsification test executable within the existing IAM MCMC infrastructure. The prediction is stated here, in advance of the result: an IAM chain with $\Omega_b h^2$ freed from its Planck prior should return a posterior consistent with $\eta \approx 6 \times 10^{-10}$ if the self-consistency loop correctly identifies the baryon asymmetry as a constrained rather than free parameter.

The broader question posed by this paper is whether the standard model's treatment of η as a free parameter is complete. Within a framework in which the universe has a finite information writing budget and that budget is connected to both the dark sector and the cosmological constant, the baryon asymmetry is not obviously free. It may be the value the universe had to have in order to become the universe it is.

It is perhaps worth noting that this reframing places the matter-antimatter asymmetry in the same conceptual category as the cosmological constant [Mahaffey, 2026d] and the virial coupling $\beta_m = \Omega_m/2$ [Mahaffey, 2026a]: quantities that appear contingent in the standard framework but emerge as necessary conditions within a framework governed by an information writing constraint. Whether this pattern reflects a deeper principle is a question the authors leave to the reader.

Data Availability

All MCMC chains, analysis scripts, and computational notebooks supporting the IAM framework are publicly available at `https://github.com/hmahaffeyges/IAM-Validation` (Zenodo DOI: 10.5281/zenodo.18702042).

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